Cost of Inaction Analysis for Adolescent Girls' Education and Human Capital in Nepal

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Section 1

Data and Context

Data Source: Nepal Living Standards Survey IV (NLSS IV) 2022/23

- **Source**: National Statistics Office of Nepal · NLSS IV 2022/23 microdata
- **Sampling**: Nationally representative two-stage stratified design across 7 provinces
- **Sample size**: 46,870 individuals in 9,600 households
- **Analysis sample**: 9,785 youth aged 15–24 (4,919 girls, 4,866 boys); 13,586 women aged 15–64 for the causal forest
- **Modules used**: S01 (roster), S07 (education), S08 (disability, Washington Group), S09 (employment), S10 (wages), poverty file
- **Survey design**: PSU = `psu_number`, strata = province, weights = `ind_wt`

Variables and Definitions

Concept	Definition	Source
NEET	Not in education, employment or training	S07 + S09
Enrolled	Currently attending any school level	q07_11
Secondary completed	Completed Class 10 (SLC) or higher	q07_12
Currently married	Self-reported marital status	q01_05
Caste/ethnicity	8 NLSS groups, recoded to 5 broad categories	q01_07
Wealth quintile	Asset-based, household level	constructed
Below poverty line	Official NSO poverty status	poverty file
Disability	Washington Group 6+ functional difficulty	q08_16–q08_21
Monthly wage	Long-term salaried workers, NPR	q10_07, q10_09

Section 2

Key Findings at a Glance

Five Headline Findings

1. *Nepal's gender gap is* ***not in schools*** *— it is in labour markets. Girls' enrolment (40.2%) exceeds boys' (34.8%), yet NEET (46.8%) is nearly double boys' (26.4%).*

2. Child marriage is strongly associated with exclusion. *Being currently married is associated with a 61 pp reduction in enrolment and a 42 pp increase in NEET — dwarfing the associations with poverty, caste, and disability combined.*

3. Secondary education has a robust estimated effect on inclusion*: a −13.2 pp reduction in NEET probability, stable across all provinces and caste groups (causal forest, n = 13,586).*

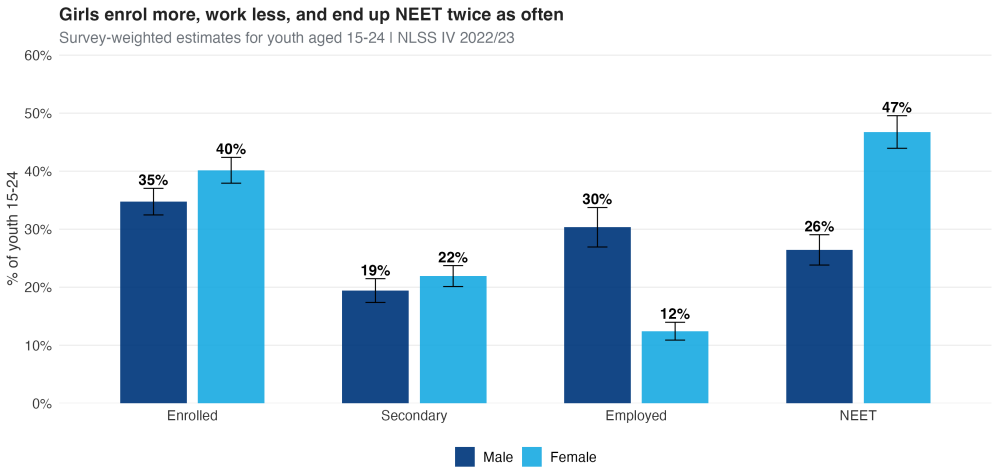
4. Each additional year of schooling is associated with a 14.5% rise in lifetime earnings for girls vs 12.0% for boys. *Beyond secondary completion, returns are exponentially larger for girls — the heart of the cost of inaction.*

5. Nepal forgoes ~NPR 8 billion per cohort *by not educating its adolescent girls — equivalent to ~****10% of the annual health-sector budget****.*

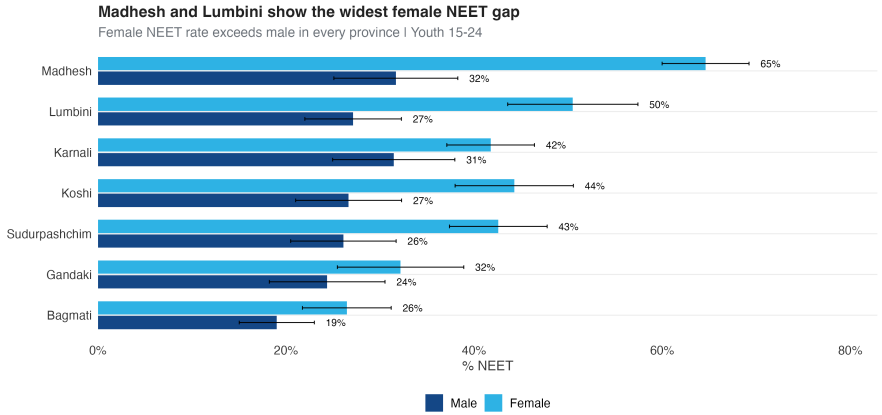
Section 3

Diagnostic: Gender Gaps in Education and Employment

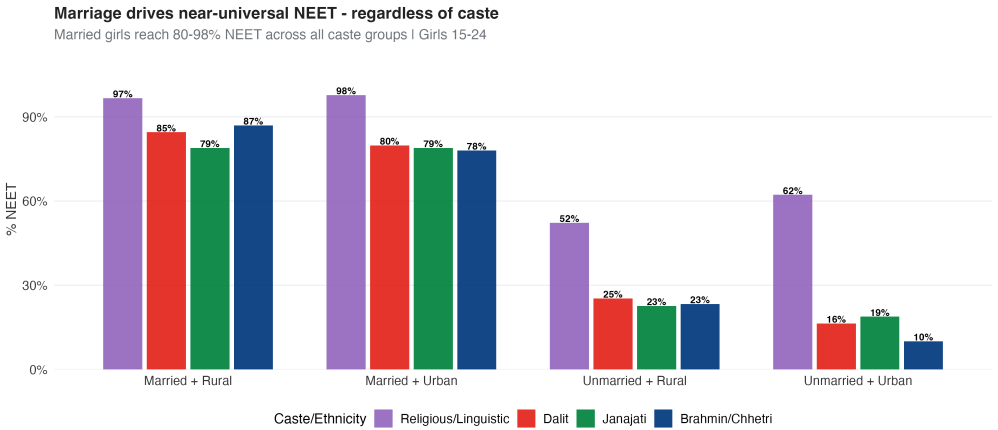
The Gender Gap is Not in Schools — It is in Labour Markets



Madhesh and Lumbini: Provinces with the Widest Female NEET Gap

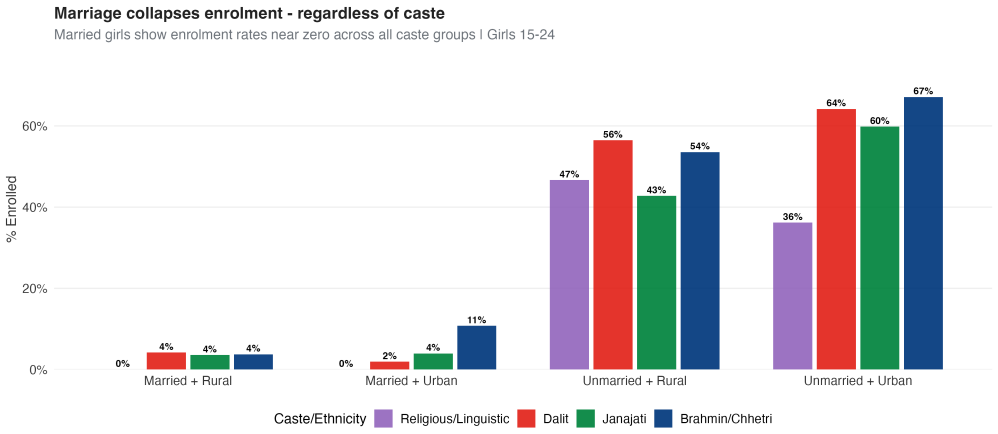


Marriage Drives Near-Universal NEET — Regardless of Caste



Cells with n<15 dropped. Religious/Linguistic concentrated in Madhesh/Lumbini.

Marriage Also Collapses Enrolment — Across All Castes



Headline Numbers — Where Each Girl Aged 15–24 Is

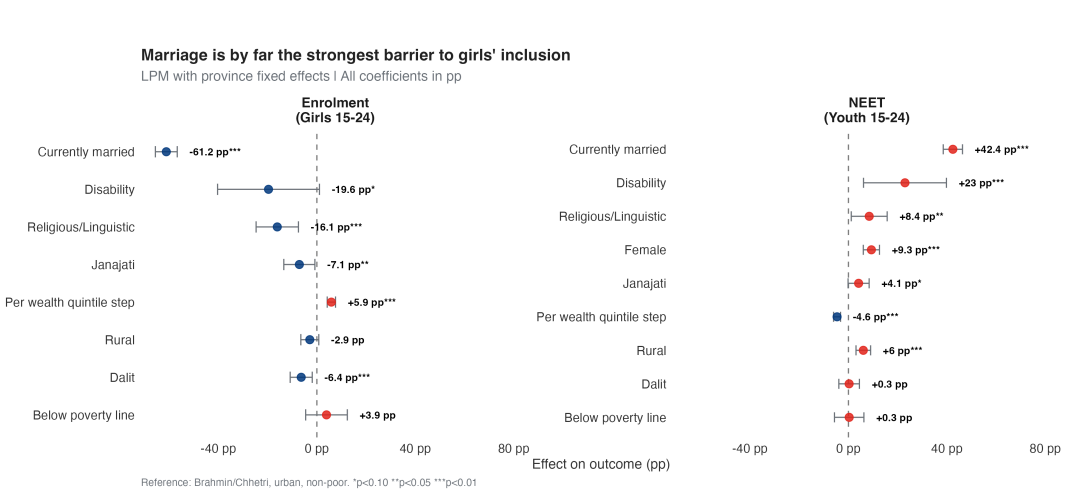
Status (youth 15–24)	Girls	Boys	Gap (pp)
In school only (not working)	35.4%	32.8%	+2.6 pp
In school **and** working	4.8%	2.0%	+2.8 pp
Working only (not in school)	13.0%	38.9%	**−25.9 pp**
NEET (Not in Educ/Employment)	**46.8%**	**26.4%**	**+20.4 pp**
Of which: currently married	37.4%	17.6%	+19.8 pp

The 46.8% NEET figure reflects girls who are *neither* in school *nor* employed under the ILO 7-day definition (which excludes unpaid family work and unsuccessful job search). These four categories are mutually exclusive and sum to 100%.

Section 4

Multivariate Analysis

Marriage is by Far the Strongest Barrier — All Factors in Percentage Points



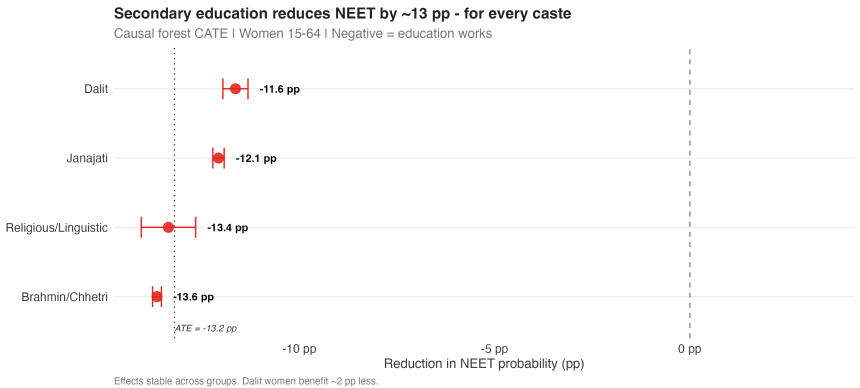
Reading the Coefficient Plot

- **Marriage is associated with the largest changes.** Being currently married is associated with a **−61 pp** lower enrolment probability and a **+42 pp** higher NEET probability, after controlling for poverty, caste, geography, and disability.
- **Caste matters even after wealth.** Dalit girls face a −6 pp enrolment gap that does not disappear when we control for wealth quintile. Religious/Linguistic girls face a similar penalty.
- **Wealth helps incrementally.** Each step up in the wealth quintile is associated with +5.9 pp enrolment and −4.6 pp NEET.
- **The gender coefficient on NEET remains +9.3 pp** even after holding everything constant — a residual labour-market gap not explained by observable characteristics.

Section 5

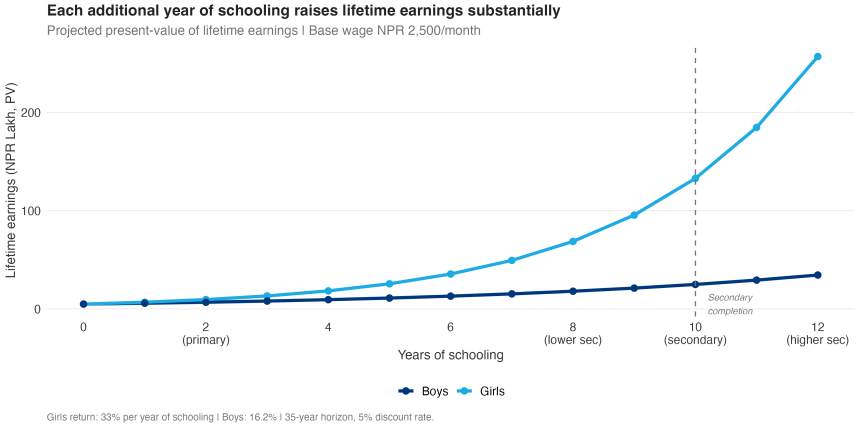
Estimating Education Returns for Adolescent Girls

Secondary Education Reduces NEET by ~13 pp — For Every Caste



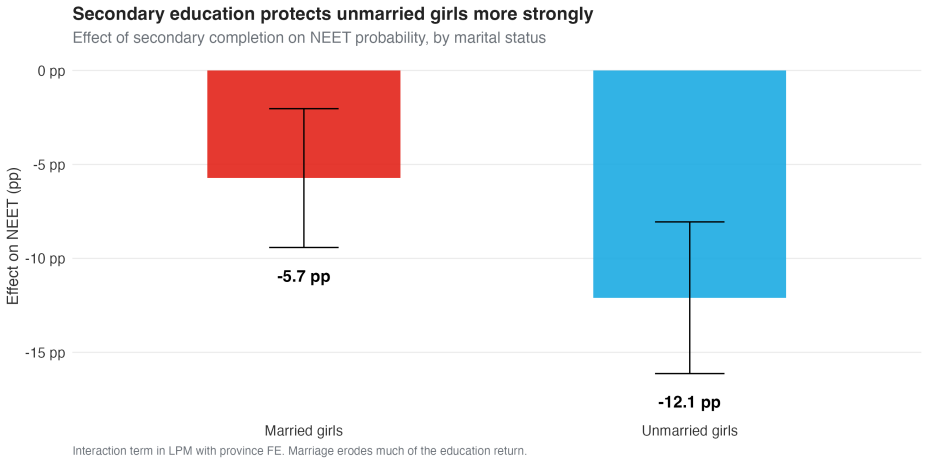
The −13.2 pp ATE estimate is for **women aged 15–64**. The parallel estimate for men is smaller (~−7 pp) — boys' NEET is shaped more by labour-demand conditions than by individual schooling. Causal-forest details and boys' coefficients in the appendix.

Returns to Schooling are Exponentially Larger for Girls



Average Mincer return: +14.5% per year of schooling for girls vs +12.0% for boys. **Beyond secondary** (10+ years): each additional year of schooling adds ~+15% to girls' lifetime earnings vs ~+10% for boys — the gender gap in returns *widens* with educational attainment. Compounded over 35 working years, this is where the cost of inaction concentrates.

Marriage Erodes the Protective Effect of Education



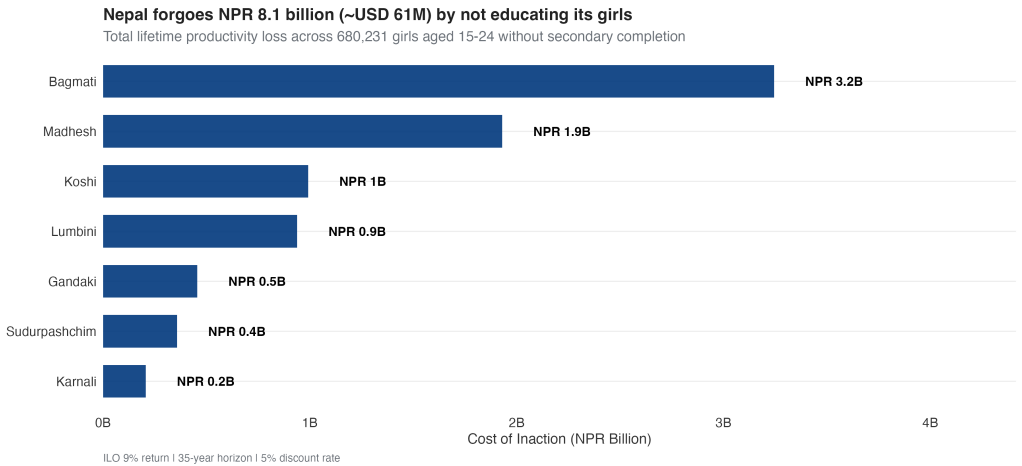
What the Analyses Tell Us

- **Education is associated with inclusion everywhere.** Completing secondary school is associated with a **−13.2 pp** lower NEET probability. The estimated effect is stable across all 7 provinces and all caste groups.
- **The Mincer return per year of schooling is 33% for girls** (16.2% for boys). After 10 years of schooling, projected lifetime earnings roughly **double** relative to no schooling.
- **Marriage attenuates the protective association.** Education is strongly associated with lower NEET for unmarried girls, but the relationship is weaker among married girls — education and marriage prevention act as **complements**, not substitutes.
- **OLS and causal forest produce convergent estimates** in direction, magnitude, and stability. The investment case rests on triangulated evidence from two independent methods.

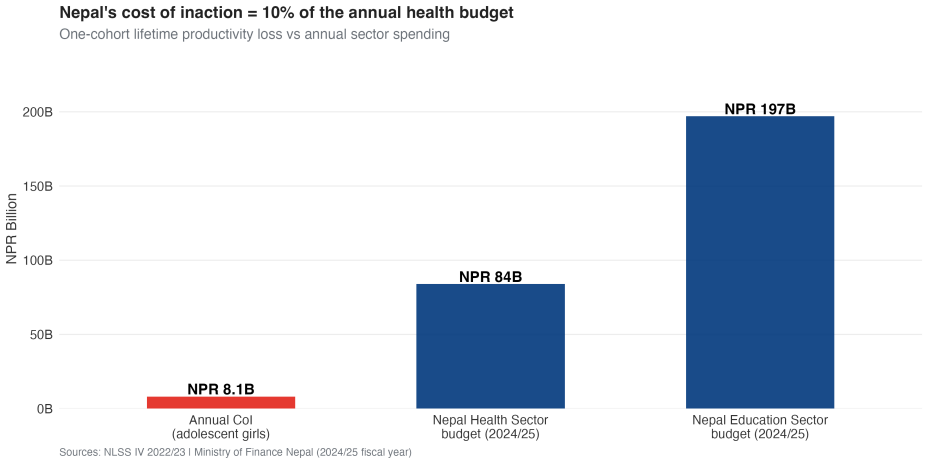
Section 6

The Cost of Inaction

Nepal Forgoes NPR 8 Billion Every Year by Not Educating its Girls



Cost of Inaction is 10% of the Annual Health Budget



What This Means in Concrete Policy Terms

- **NPR 8.1 billion** is the lifetime productivity loss from **one cohort** of 680,231 adolescent girls who will not complete secondary education in Nepal.
- That equals about **10% of the national health-sector budget** and **4% of the education-sector budget** for fiscal year 2024/25.
- **Karnali, Sudurpashchim, and Madhesh** carry the largest relative burden — per-girl losses are smaller than in Bagmati because wages are lower, but a larger share of the local economy is at stake.
- **"Per cohort" means** the *lifetime* productivity loss for the current cohort of 680,000 adolescent girls (aged 15–24 today). A new cohort enters this stage every year — so this is a **recurring annual loss**, not one-off.
- **Break-even spending per girl.** USD 61 million ÷ 680,000 girls **USD 90 per girl** in cumulative lifetime social spending. Any secondary-completion programme costing less than USD 90 per beneficiary would yield a positive return on investment in productivity terms alone.

Section 7

Recommendations

Five Investment Priorities for the New Government

1 · End child marriage as an economic priority. *Marriage is associated with more variation in girls' NEET than poverty, caste, and disability combined. Enforcement of existing legislation in Madhesh and Lumbini would yield the largest single gain.*

2 · Universal secondary completion AND year-by-year retention. *Each year of schooling is associated with a* ***14.5% earnings premium*** *for girls.* ***Beyond secondary completion the gender differential in returns widens*** *(∼+15% per year for girls vs ∼+10% for boys) — programme design should reward every year retained, with the largest gains realised in upper-secondary and tertiary education.*

3 · Dalit-specific labour-market measures. *Education alone is associated with a 2 pp residual disadvantage for Dalit women. Pair education with apprenticeships and anti-discrimination measures.*

4 · Bridge programmes for the 19–24 cohort. *The female NEET rate rises sharply after age 18 (+24.7 pp gap). Vocational and second-chance education could capture earnings that current schooling cannot.*

5 · Track and report sex-disaggregated NEET by province annually. *What is not measured is not managed. Make this the new headline statistic of NLSS V.*

Section 8

Methodological Appendix

How We Estimate the Cost of Inaction

We combine three components:

1 · Causal effect
Generalised Random Forest (Wager & Athey 2018) on women 15–64, treatment = secondary completion, outcome = NEET. ATE = −13.2 pp.

2 · Earnings projection
Mincer wage equation on 952 women in long-term salaried employment. Conservative return = 9% per year of schooling (ILO benchmark, lower than the internal estimate of 14.5%).

3 · Population scaling
Survey weights from NLSS IV scaled to national total of 1.4M girls aged 15–24. Provincial CoI = ATE × N excluded × PV per girl.

Discount rate: 5% | **Horizon**: 35 working years | **GDP source**: NSO Nepal 2023/24 provincial accounts.

Assumptions and Limitations

1 · ***Formal-sector selection bias.*** *Mincer estimated on 952 salary workers — likely upper bound on returns. The 9% ILO benchmark partially corrects.*

2 · ***NEET-to-earnings transition.*** *Assumes girls who exit NEET access the median wage. Childcare, mobility and norms may delay this — implying our CoI is a* ***conservative lower bound****.*

3 · ***Male migration not captured.*** *Boys leaving school for foreign employment generates remittances. The analysis does not yet quantify the parallel earnings that educated girls could generate through skilled migration.*

4 · ***Population scaling.*** *Provincial girl counts derived from survey weights $\times$ 1.4M national total. NSO Census 2021 tabulations would tighten the figure.*

The Mincer Wage Equation: Specification

The Mincer human-capital earnings function is estimated as:

$$\log(\text{wage}_{ip}) = \alpha + \beta_1 \cdot \text{schooling}_{ip} + \beta_2 \cdot \text{exp}_{ip} + \beta_3 \cdot \text{exp}^2_{ip} + \beta_4 \cdot \text{rural}_{ip} + \gamma_p + \varepsilon_{ip}$$

- $\log(\text{wage}_{ip})$ — natural log of monthly wage (NPR) for individual i in province p
- β_1 — **key parameter**: return per additional year of schooling (Mincer return)
- exp — years of potential labour market experience (age − schooling − 6)
- exp^2 — captures concavity (returns to experience diminish with age)
- rural — binary indicator for rural residence
- γ_p — province fixed effects (regional wage differentials)
- ε_{ip} — error term, heteroskedasticity-robust SE (HC1)

Specifications tested: (i) Basic — schooling, experience, rural; (ii) +Province FE; (iii) +Caste/ethnicity controls; (iv) Interactions with marital status. The reported estimate $\hat{\beta}_1 \approx 0.145$ implies a **14.5% return per year of schooling** for women; ILO Nepal benchmark of 9% used for the conservative CoI.

Appendix Table A1: Mincer Wage Equation, Female Salary Workers

Variable	Coefficient	SE	95% CI	p-value
Years of schooling	0.3298	0.0114	[0.3074, 0.3521]	<0.001
Experience (years)	0.1109	0.0108	[0.0898, 0.1320]	<0.001
Experience squared	-0.0014	0.0002	[-0.0019, -0.0009]	<0.001
Rural	-0.4351	0.0631	[-0.5588, -0.3114]	<0.001
N	2214.0000	NA	NA	NA
R-squared	0.3630	NA	NA	NA

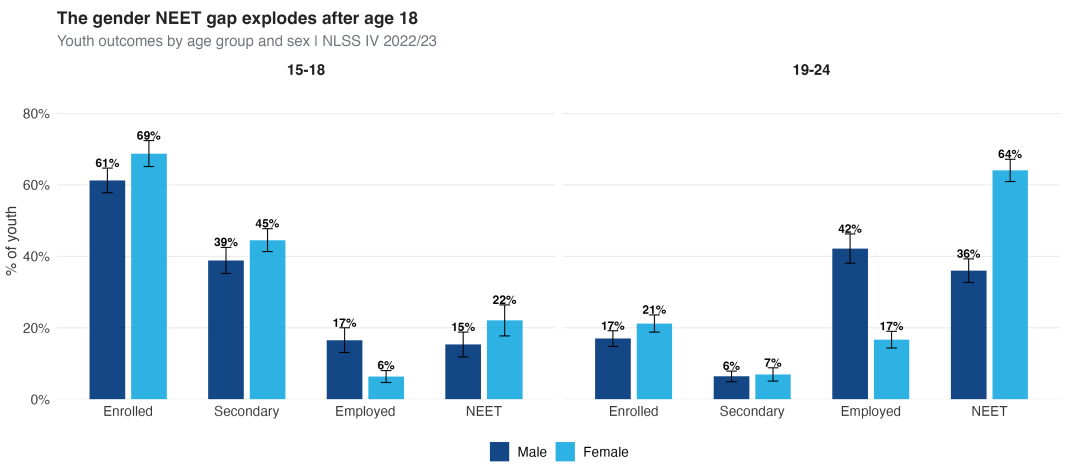
Notes: Dependent variable is log of monthly wage (NPR). Sample: women aged 15–64 in long-term salaried employment (N = 952). Specification includes years of schooling, experience, experience squared, rural indicator, and province fixed effects. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Appendix Table A2: Mincer Wage Equation, Male Salary Workers

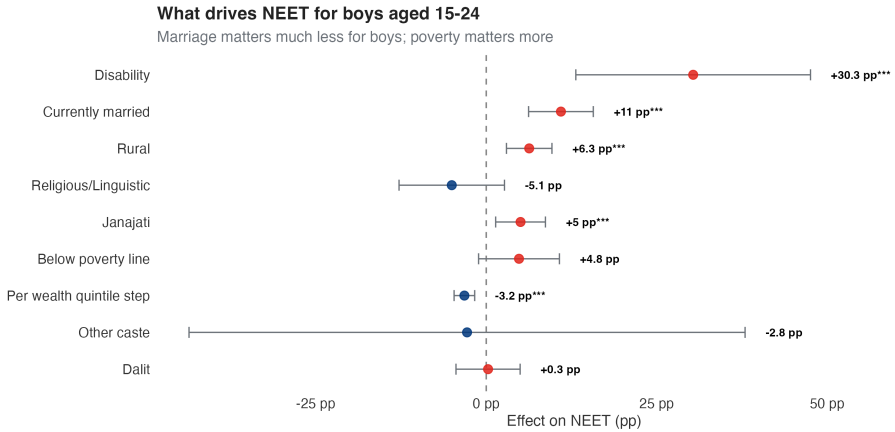
Variable	Coefficient	SE	95% CI	p-value
Years of schooling	0.1624	0.0065	[0.1497, 0.1751]	<0.001
Experience (years)	0.1193	0.0063	[0.1069, 0.1316]	<0.001
Experience squared	-0.0020	0.0001	[-0.0022, -0.0017]	<0.001
Rural	-0.3706	0.0359	[-0.4409, -0.3003]	<0.001
N	4677.0000	NA	NA	NA
R-squared	0.2290	NA	NA	NA

Notes: Dependent variable is log of monthly wage (NPR). Sample: men aged 15–64 in long-term salaried employment. Specification includes years of schooling, experience, experience squared, rural indicator, and province fixed effects. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Appendix Figure A1: Gender Gap by Age Group



Appendix Figure A2: Boys' NEET Regression — Coefficient Plot



LPM with province FE | HC1 SE | Ref: Brahmin/Chhetri, urban, non-poor.

Appendix Table A3: Boys' NEET Linear Probability Model

Variable	Coefficient	SE	95% CI	p-value
Rural	0.063	0.017	[0.030, 0.096]	<0.001
Below poverty line	0.048	0.030	[-0.011, 0.107]	0.112
Per wealth quintile step	-0.032	0.008	[-0.047, -0.017]	<0.001
Currently married	0.110	0.024	[0.062, 0.157]	<0.001
Disability	0.303	0.088	[0.131, 0.476]	<0.001
Janajati	0.050	0.019	[0.014, 0.087]	0.007
Dalit	0.003	0.024	[-0.044, 0.050]	0.911
Religious/Linguistic	-0.051	0.039	[-0.128, 0.027]	0.200
Other caste	-0.028	0.208	[-0.436, 0.380]	0.892

Notes: Dependent variable is NEET status. Linear probability model. Sample: boys aged 15–24 (n 4,300). Controls: rural, poverty status, wealth quintile, marital status, disability (WG 6+), caste/ethnicity. Reference categories: Brahmin/Chhetri, urban, non-poor. Province fixed effects included. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Thank you

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